MATHEMATICAL RESEARCH DOSSIER

Joint Ternary-Block Layer-Wise Reconstruction for 100B Donor Models

A formal treatment of the conjunctive 1.58-bit quantization and 48 KB contiguous block-sparsity problem under Hessian-driven layer-wise feature matching — solving the joint ADMM / OBC / proximal formulation, with theoretical quality-vs-sparsity degradation bounds, progressive block-wise healing, and the concrete T4 implementation pipeline.

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PROJECT

SiliconLLM Engine Co-Design Architecture

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Contents

1 Introduction and Established Experimental Laws

The SiliconLLM Engine Co-Design Architecture project has, through a sustained sequence of empirical probes on 100B-parameter donor models, established three physical and mathematical facts that bound any feasible compression strategy. These facts are not heuristic observations but hard constraints: any algorithm that violates them degrades the donor model irrecoverably or yields zero wall-clock benefit on the target C SIMD engine. We open this dossier by enumerating them in the form that will be referenced throughout the remainder of the document.

D2 Falsification — Weight Rotations Are Dead. Block-Hadamard, Cayley, and dense random-orthogonal rotations applied to *weight* matrices push the empirical weight distribution toward the i.i.d. Gaussian limit, raising the 99th-percentile mass ratio $\text{frac}_{99} \approx 0.7345$ from the native coordinate-concentrated value $\text{frac}_{99} \approx 0.64$. This destroys the very coordinate concentration that any structured-pruning or ternarization scheme exploits, and consequently *rotations of weights are forbidden as a sparsity-inducing mechanism.* Rotations remain admissible on the *activation* side (D2b), where they serve skip-gating and probing purposes, but they are categorically excluded from the weight-side optimization considered here.

The ρ-Law of CPU Granularity (48 KB Floor). The C SIMD engine is structurally unable to extract speed from scattered zeros. At the target 1.58-bit ternary density (0.5 B/weight), a single ρ-safe contiguous memory read of 48 KB corresponds to exactly 21.3 consecutive co-active SwiGLU neurons (64 when organs are separated). Any sparsity pattern below this granularity is invisible to the runtime: unstructured sparsity and the widely-deployed 2:4 semi-structured pattern both yield a block-skippable fraction of approximately 0.001 on the C engine, i.e. they contribute *zero* speedup. We therefore treat all zero patterns that do not aggregate into ≥ 48 KB contiguous blocks as computationally inert and constrain our search to block-sparsity at this granularity or coarser.

Layer-Wise Feature-Matching Objective (1000× Cheaper Than End-to-End). The donor model provides, at no additional supervision cost, a D-dimensional output feature vector at every position and every layer. We do not retrain the donor under next-token cross-entropy; instead we solve, per layer, the reconstruction problem

$$\min_{W'} \; \mathbb{E}_X \Big[\, \| W'X - WX \|_F^2 \, \Big] \;=\; \min_{W'} \; \mathrm{Tr}\Big((W' - W)\, H\, (W' - W)^\top \Big) \,,$$

where $H = XX^\top \in \mathbb{R}^{N \times N}$ is the uncentered activation covariance (Hessian). For a 100B donor, the full layer-wise reconstruction over calibration activation tensors X costs approximately 13 GPU-hours on a single NVIDIA T4, reducing compute by three orders of magnitude relative to end-to-end fine-tuning.

These three laws together imply a sharp desideratum: we require a solver that simultaneously respects (i) the contiguous block granularity ≥ 48 KB, (ii) the ternary grid $\{-S, 0, +S\}$ with per-block or per-channel FP32 scaling, and (iii) the in-place predictable activation-sparsity needed by the SIMD engine's `pshufb` lookup-table probe. The existing literature — SparseGPT [1], Wanda [2], SmoothQuant [3], SpinQuant [4] — operates sequentially: prune, then quantize, then hope for contiguity. Sequential operations do not compose. The remainder of this dossier is devoted to formulating, analyzing, and engineering a *joint* solver on the conjoint ternary-block manifold.

2 Problem Statement and Mathematical Formulation

2.1 Notation and Setup

Let the donor model possess L transformer layers (typically $L \geq 80$ for a 100B donor). For each layer $\ell \in \{1, \ldots, L\}$, let the weight matrix under consideration be $W_\ell \in \mathbb{R}^{M \times N}$, where M denotes the output feature dimension and N the input feature dimension. We will suppress the layer subscript ℓ whenever a single-layer analysis is in progress. The calibration activation tensor at layer ℓ is $X_\ell \in \mathbb{R}^{N \times B_{\text{tokens}}}$, where B_{tokens} is the total number of calibration tokens (typically $B_{\text{tokens}} \in [2048, 8192]$). The activation covariance (Hessian) is

$$H \;\coloneqq\; XX^\top \;\in\; \mathbb{R}^{N \times N},$$

which is symmetric positive semidefinite. Under the standard assumption that calibration activations are full-rank in feature space (which holds for $B_{\text{tokens}} \geq N$), H is positive definite on a low-rank subspace of dimension $r = \operatorname{rank}(H) \leq \min(N, B_{\text{tokens}})$.

The donor model's per-layer output on the calibration set is the reference signal $Y = WX \in \mathbb{R}^{M \times B_{\text{tokens}}}$. Any compressed weight W' induces an output $Y' = W'X$, and the layer-wise reconstruction error is

$$\mathcal{E}(W') \;\coloneqq\; \|Y' - Y\|_F^2 \;=\; \operatorname{Tr}\Big((W' - W)\, H\, (W' - W)^\top\Big). \tag{1}$$

The right-hand side decomposes the error into a quadratic form in $W' - W$ weighted by H, which is the canonical layer-wise objective studied throughout the quantization literature [1, 11]. The naive objective (1) is solved trivially by $W' = W$; the difficulty arises entirely from the constraint set, which we now formalize.

2.2 The Joint Feasible Manifold

Definition 2.1 (Ternary-Block Joint Manifold)**.** Fix a block partition of the rows of W into contiguous groups $\mathcal{B} = \{\mathcal{B}_1, \mathcal{B}_2, \ldots, \mathcal{B}_K\}$, where each block $\mathcal{B}_k \subseteq \{1, \ldots, M\}$ is a contiguous index set with $|\mathcal{B}_k| = B_{\text{block}} \geq 21.3$ (the ρ-safe granularity). Let $\mathbf{s} \in \{0,1\}^K$ be a binary block-support vector, with $\mathbf{s}_k = 1$ indicating block k is *active* (retained) and $\mathbf{s}_k = 0$ indicating it is *dropped*. Let $S \in \mathbb{R}_+^K$ be per-block FP32 scaling factors. The joint feasible manifold is

$$\mathcal{M} \;\coloneqq\; \left\{ W' \in \mathbb{R}^{M \times N} \;\middle|\; \begin{array}{l} W'_{\mathcal{B}_k,\cdot} \in \{-S_k, 0, +S_k\}^{B_{\text{block}} \times N} \text{ for all active blocks } k, \\ W'_{\mathcal{B}_k,\cdot} = 0 \text{ for all inactive blocks } k \text{ with } \mathbf{s}_k = 0, \\ \mathbf{s} \text{ linearly probe-predictable} \\ \quad \text{on the input activation stream} \end{array} \right\}.$$

Three remarks on Definition 2.1 clarify its physical meaning.

Remark 2.2 (Block Coarsening vs. Unstructured Sparsity)**.** The manifold $\mathcal{M}$ is non-convex, disconnected, and of combinatorial cardinality. The constraint that the zero pattern be organized in blocks of size ≥ 21.3 neurons is what converts an unstructured sparsity problem (combinatorial over $\binom{MN}{s}$ choices) into a block-sparsity problem (combinatorial over $\binom{K}{K - K \cdot s}$ choices, exponentially smaller when $B_{\text{block}} \gg 1$). This coarsening is what makes the C SIMD engine able to skip 48 KB memory regions wholesale via `pshufb` mask precomputation.

Remark 2.3 (Per-Block vs. Per-Channel Scaling)**.** The ternary grid $\{-S_k, 0, +S_k\}$ uses per-block scale S_k, which is the natural choice when blocks are co-active: a single FP32 scale is

loaded once per block, amortized over $B_{\text{block}} \cdot N$ weights. Per-channel scaling (one S_m per output row) would require M scales, defeating the cache locality gain that motivates the block structure in the first place.

Remark 2.4 (Probe-Alignment Constraint)**.** The third constraint — that the active block pattern $\mathbf{s}$ be predictable by a linear probe on the input stream — is what enables in-place zero-skip execution. Concretely, the C engine computes a probe vector $p = \text{sign}(Ax_{\text{in}}) \in \{-1, +1\}^K$ from a small $K \times N$ matrix A (the probe), and uses `pshufb` to skip the load and compute for blocks where $p_k = -1$. This requires the active set to align with the half-space partition induced by the probe, which is a non-trivial coupling between W' and the probe geometry.

2.3 The Joint Optimization Problem

With Definition 2.1 in hand, the central problem of this dossier is:

$$\boxed{W'^{\star} \;=\; \underset{W' \in \mathcal{M}}{\text{argmin}}\; \text{Tr}\Big((W' - W)\, H\, (W' - W)^{\top}\Big)} \tag{2}$$

Problem (2) is the formal object that the remainder of this dossier addresses. Section 3 develops an ADMM-based solver with provable convergence on $\mathcal{M}$. Section 4 derives the theoretical quality-vs-sparsity degradation bounds, comparing contiguous-block, unstructured, and 2:4 sparsity regimes. Section 5 addresses the propagation of per-layer errors across the 80+ layers of a 100B donor. Section 6 specifies the concrete PyTorch/CUDA implementation that achieves the $\sim$ 10–15 minute per-layer budget on a single T4 GPU.

2.4 Decomposition into Block Sub-Problems

A key structural observation, which we shall exploit repeatedly, is that under the block partition $\mathcal{B}$ the Hessian-weighted objective (1) decomposes across blocks *only when H is block-diagonal*, which is generically not the case. However, the objective decomposes across blocks in the sense that the per-block sub-matrix $H_{\mathcal{B}_k,\mathcal{B}_k}$ captures the dominant intra-block coupling, while inter-block coupling appears through off-diagonal blocks $H_{\mathcal{B}_k,\mathcal{B}_j}$ for $k \neq j$. We exploit this by partitioning

$$H \;=\; \Big(H_{kk'}\Big)_{k,k'=1}^{K}, \qquad H_{kk'} \in \mathbb{R}^{B_{\text{block}} \times B_{\text{block}}},$$

and solving a sequence of block-local problems that approximate the joint one via a Schur-complement correction (developed in Section 3). The per-block sub-problem, which forms the inner loop of our ADMM iteration, is:

$$W'^{\star}_{\mathcal{B}_k} \;=\; \underset{W'_{\mathcal{B}_k} \in \{-S_k, 0, +S_k\}^{B_{\text{block}} \times N}}{\text{argmin}}\; \text{Tr}\Big((W'_{\mathcal{B}_k} - W_{\mathcal{B}_k})\, H_{kk}\, (W'_{\mathcal{B}_k} - W_{\mathcal{B}_k})^{\top}\Big) \;+\; \text{coupling correction}. \tag{3}$$

The coupling correction is the source of the inter-block dependence; in the limit of weak coupling (small off-diagonal blocks), the per-block problems decouple and the solver reduces to a parallel block-ternarization. We quantify this decoupling precisely in Section 4.

3 Part A — Algorithmic Formulation of the Joint Solver

We attack Problem (2) by splitting the conjoint constraint set $\mathcal{M} = \mathcal{T}_{\text{ternary}} \cap \mathcal{T}_{\text{block}} \cap \mathcal{T}_{\text{probe}}$ across an Alternating Direction Method of Multipliers (ADMM) iteration. The three constraints — ternary quantization, contiguous block support, and probe alignment — each define a projection

operator with a closed-form or near-closed-form evaluation, while the Hessian-weighted fidelity defines the smooth quadratic term. ADMM is the canonical framework for problems of this shape: a smooth convex quadratic coupled to non-convex projection operators via a splitting variable [5, 6].

3.1 ADMM Splitting of the Joint Problem

Introduce splitting variables $Z_1, Z_2, Z_3 \in \mathbb{R}^{M\times N}$ and rewrite Problem (2) as

$$\min_{W',Z_1,Z_2,Z_3} \operatorname{Tr}\Big((W'-W)\,H\,(W'-W)^\top\Big) \;+\; \mathcal{I}_{\mathcal{T}_{\text{ternary}}}(Z_1) \;+\; \mathcal{I}_{\mathcal{T}_{\text{block}}}(Z_2) \;+\; \mathcal{I}_{\mathcal{T}_{\text{probe}}}(Z_3) \tag{4}$$

subject to consensus constraints $W' = Z_1 = Z_2 = Z_3$, where $\mathcal{I}_{\mathcal{C}}$ is the indicator function (zero on $\mathcal{C}$, $+\infty$ elsewhere). The augmented Lagrangian of (4), scaled-form ADMM [5], is

$$\begin{aligned}\mathcal{L}_\rho(W', Z_1, Z_2, Z_3, U_1, U_2, U_3) \;=\; &\operatorname{Tr}\Big((W'-W)\,H\,(W'-W)^\top\Big) \\ &+ \sum_{i=1}^{3}\Big[\,\mathcal{I}_{\mathcal{C}_i}(Z_i) + \tfrac{\rho_i}{2}\,\|W' - Z_i + U_i\|_F^2\,\Big],\end{aligned} \tag{5}$$

where $\rho_1, \rho_2, \rho_3 > 0$ are per-constraint penalty parameters and U_i are scaled dual variables. The ADMM iteration consists of alternating minimizations of (5) with respect to each primal variable, followed by dual ascent.

3.2 ADMM Update Equations

The four blocks of updates, executed in sequence per outer iteration $t = 0, 1, 2, \ldots$, are:

(W1) W'-update (smooth quadratic). Holding $Z_i^{(t)}$ and $U_i^{(t)}$ fixed, the W'-minimization is a convex quadratic in W':

$$W'^{(t+1)} \;=\; \operatorname*{argmin}_{W'} \operatorname{Tr}\Big((W'-W)\,H\,(W'-W)^\top\Big) + \sum_{i=1}^{3}\tfrac{\rho_i}{2}\left\|W' - Z_i^{(t)} + U_i^{(t)}\right\|_F^2. \tag{6}$$

Stationarity yields the linear system (vectorizing for clarity, with $\operatorname{vec}(W') \in \mathbb{R}^{MN}$):

$$\Big(H \otimes I_M + \Big(\textstyle\sum_{i=1}^{3}\rho_i\Big)\,I_{MN}\Big)\,\operatorname{vec}(W'^{(t+1)}) \;=\; \operatorname{vec}\Big(WH + \textstyle\sum_{i=1}^{3}\rho_i\left(Z_i^{(t)} - U_i^{(t)}\right)\Big). \tag{7}$$

Crucially, the system matrix $H \otimes I_M + \rho_\Sigma I_{MN}$, with $\rho_\Sigma = \sum_i \rho_i$, is block-separable across the M output rows. Each row m admits an independent solve of an $N \times N$ linear system

$$(H + \rho_\Sigma I_N)\;W'^{(t+1)\top}_{m,\cdot} \;=\; (WH)^\top_{m,\cdot} + \rho_\Sigma\left(Z^{(t)} - U^{(t)}\right)^\top_{m,\cdot}. \tag{8}$$

The matrix $H + \rho_\Sigma I_N$ is symmetric positive definite whenever $H \succeq 0$ and $\rho_\Sigma > 0$, which permits a single Cholesky factorization $H + \rho_\Sigma I_N = LL^\top$ at the start of ADMM (cost $\mathcal{O}(N^3)$, paid once per layer) followed by $\mathcal{O}(N^2)$ back-substitutions per row per iteration (cost $\mathcal{O}(MN^2)$ per iteration). On a T4 GPU the Cholesky is dispatched to `torch.linalg.cholesky` with TF32 enabled.

(Z1) Ternary Projection. With $W'^{(t+1)}$ in hand, the Z_1-update is the projection onto the ternary manifold:

$$Z_1^{(t+1)} \;=\; \Pi_{\mathcal{T}_{\text{ternary}}}\Big(W'^{(t+1)} + U_1^{(t)}\Big), \tag{9}$$

where, for each block k with scale S_k, the projection operator is applied entry-wise to the matrix argument $A = W'^{(t+1)} + U_1^{(t)}$ on block $\mathcal{B}_k$:

$$\left[\Pi_{\mathcal{T}_{\text{ternary}}}(A)\right]_{\mathcal{B}_k,\cdot} \;=\; S_k \cdot \operatorname{sign}(A_{\mathcal{B}_k,\cdot}) \odot \mathbb{1}[\,|A_{\mathcal{B}_k,\cdot}| \geq \tau_k\,], \tag{10}$$

with $\operatorname{sign}(0) = 0$ by convention. The per-block scale S_k is chosen to minimize the per-block quantization error in closed form:

$$S_k \;=\; \frac{\sum_{(m,n)\in\mathcal{B}_k\times[N]} |W_{m,n}|\,\mathbb{1}[|W_{m,n}| \geq \tau_k]}{|\{(m,n) \in \mathcal{B}_k \times [N] : |W_{m,n}| \geq \tau_k\}|}, \tag{11}$$

i.e. the mean of the magnitudes that exceed the threshold. The threshold τ_k is set so that exactly the fraction $\phi_{\text{ternary}} \in [0.5, 0.95]$ of largest-magnitude entries in block k survive quantization; this controls the within-block density.

(Z2) Contiguous Block-Support Projection. The Z_2-update enforces the contiguous block-sparsity constraint. We do not project $W'^{(t+1)} + U_2^{(t)}$ entry-wise onto a sparse pattern; rather, we select which *blocks* are active:

$$Z_2^{(t+1)} \;=\; \Pi_{\mathcal{T}_{\text{block}}}\Big(W'^{(t+1)} + U_2^{(t)}\Big), \tag{12}$$

where the block-support projection proceeds in two stages. *Stage 1 (per-block energy scoring).* For each block k, compute the Hessian-weighted block energy

$$E_k \;=\; \operatorname{Tr}\Big(W_{\mathcal{B}_k,\cdot}\, H_{kk}\, W_{\mathcal{B}_k,\cdot}^{\top}\Big) \;+\; \lambda_{\text{off}} \sum_{j\neq k} \operatorname{Tr}\Big(W_{\mathcal{B}_k,\cdot}\, H_{kj}\, W_{\mathcal{B}_j,\cdot}^{\top}\Big), \tag{13}$$

which approximates the contribution of block k to the reconstruction error if block k is dropped. The first term is the block-self energy; the second is a regularizer capturing inter-block coupling, weighted by $\lambda_{\text{off}} \in [0, 0.1]$. *Stage 2 (greedy block selection).* Sort blocks by E_k descending and mark the top $K_{\text{active}} = \lceil (1-s)\cdot K \rceil$ blocks as active, where s is the target block-sparsity. Zero out the rest:

$$(Z_2^{(t+1)})_{\mathcal{B}_k,\cdot} \;=\; \begin{cases} (W'^{(t+1)})_{\mathcal{B}_k,\cdot} + (U_2^{(t)})_{\mathcal{B}_k,\cdot} & \text{if } E_k \text{ in top-}K_{\text{active}}, \\ 0 & \text{otherwise.} \end{cases} \tag{14}$$

(Z3) Probe-Alignment Projection. The Z_3-update aligns the block-support pattern with the linear activation probe. Let $A \in \mathbb{R}^{K\times N}$ be the probe matrix (which may itself be optimized, but is held fixed during a single ADMM solve). The probe predicts, for each input activation $x \in \mathbb{R}^N$, a binary active-set indicator $p(x) = \operatorname{sign}(Ax) \in \{-1, +1\}^K$. We require the active-block pattern at calibration token j, denoted $\mathbf{s}^{(j)} \in \{0,1\}^K$, to satisfy $p(x_j)_k = +1 \Rightarrow \mathbf{s}_k^{(j)} = 1$ (probe-positive blocks must be active). The projection is:

$$(Z_3^{(t+1)})_{\mathcal{B}_k,\cdot} \;=\; \begin{cases} (W'^{(t+1)})_{\mathcal{B}_k,\cdot} + (U_3^{(t)})_{\mathcal{B}_k,\cdot} & \text{if } p(x_j)_k = +1 \text{ for some } j \in [B_{\text{tokens}}], \\ 0 & \text{if } p(x_j)_k = -1 \text{ for all } j \in [B_{\text{tokens}}]. \end{cases} \tag{15}$$

The probe matrix A is updated periodically (every $T_{\text{probe}} = 50$ ADMM iterations) by solving the small logistic-regression problem

$$A^\star \;=\; \operatorname*{argmin}_{A\in\mathbb{R}^{K\times N}} \sum_{j=1}^{B_{\text{tokens}}} \sum_{k=1}^{K} \log\Big(1+\exp\Big(-p_k^{(j)}(A)\,(2\mathbf{s}_k^{(j)}-1)\Big)\Big)\,, \tag{16}$$

where $p_k^{(j)}(A) = \operatorname{sign}((Ax_j)_k)$ and $\mathbf{s}^{(j)}$ is the block-support pattern obtained from the previous Z_2-update. This sub-problem is a smooth convex logistic regression and converges in ~ 100 steps of L-BFGS.

(U) Dual Updates. The dual variables are updated by standard dual ascent:

$$U_i^{(t+1)} \;\leftarrow\; U_i^{(t)} + W'^{(t+1)} - Z_i^{(t+1)}, \qquad i\in\{1,2,3\}. \tag{17}$$

3.3 Convergence Analysis

The joint problem (2) is non-convex owing to the ternary grid and the combinatorial block-support selection. Classical ADMM convergence theory [5] therefore does not apply directly. We establish convergence under the following assumptions.

Assumption 3.1 (Restricted Strong Convexity of H)**.** The Hessian H satisfies a restricted strong convexity (RSC) condition: there exists $\mu > 0$ such that for all $\Delta \in \mathbb{R}^{M\times N}$ whose row-support lies in a union of at most K_{active} blocks (i.e. Δ is block-sparse with the same support as the ADMM iterate),

$$\operatorname{Tr}\Big(\Delta H \Delta^\top\Big) \;\geq\; \mu\,\|\Delta\|_F^2\,. \tag{18}$$

Assumption 3.2 (Lipschitz Projections)**.** The projection operators $\Pi_{\mathcal{T}_{\text{ternary}}}$, $\Pi_{\mathcal{T}_{\text{block}}}$, $\Pi_{\mathcal{T}_{\text{probe}}}$ are Lipschitz-continuous on a bounded neighborhood of the iterates, with Lipschitz constants $L_1, L_2, L_3 \leq 1$ (projections onto non-expansive sets are non-expansive; the ternary projection (10) is in fact 1-Lipschitz as it is a proximal operator).

Theorem 3.3 (Ergodic Convergence of Joint ADMM)**.** *Under Assumptions 3.1–3.2, if the penalty parameters satisfy $\rho_\Sigma > \mu$ and the sequence of iterates $\{(W'^{(t)}, Z_i^{(t)}, U_i^{(t)})\}$ remains bounded (which holds for any finite T given a finite initialization), then the ergodic average $\bar{W}'^{(T)} = \frac{1}{T}\sum_{t=1}^{T} W'^{(t)}$ satisfies*

$$\mathcal{E}(\bar{W}'^{(T)}) - \mathcal{E}(W'^\star) \;\leq\; \frac{C_1}{T}\left(\left\|W'^{(0)} - W'^\star\right\|_F^2 + \sum_{i=1}^{3}\frac{1}{\rho_i}\left\|U_i^{(0)} - U_i^\star\right\|_F^2\right) \;+\; C_2\,\delta_{proj}, \tag{19}$$

where $C_1 = \mathcal{O}(\mu^{-1}(1+L_{\max}^2))$, $L_{\max} = \max_i L_i$, $C_2 = \mathcal{O}(\lambda_{off})$, and $\delta_{proj} \geq 0$ is the residual consensus violation $\max_i \|W' - Z_i\|_F$. Moreover, $\delta_{proj} \to 0$ as $t \to \infty$ provided ρ_i are chosen such that the augmented Lagrangian satisfies the Kurdyka–Łojasiewicz inequality on the joint manifold.

Proof sketch. The proof proceeds in three stages. *(i) Sufficient descent.* Each primal update decreases the augmented Lagrangian $\mathcal{L}_\rho$ by at least $\frac{\mu+\rho_\Sigma}{2}\left\|W'^{(t+1)} - W'^{(t)}\right\|_F^2$ (using the RSC inequality (18)). *(ii) Bounded dual residual.* The dual update (17) telescopes, giving $\left\|U_i^{(t)}\right\|_F \leq \left\|U_i^{(0)}\right\|_F + \sum_{\tau=0}^{t-1}\left\|W'^{(\tau+1)} - Z_i^{(\tau+1)}\right\|_F$, which is bounded because the primal updates are bounded (under the bounded-iterate assumption). *(iii) KL inequality.* The augmented Lagrangian $\mathcal{L}_\rho$ is

a semi-algebraic function (its components are polynomial and indicator-of-semi-algebraic-sets), hence satisfies the Kurdyka–Łojasiewicz inequality [7, 8]. By the standard KL-based convergence argument for non-convex ADMM [6], the iterates converge to a critical point of $\mathcal{L}_\rho$, and the rate (19) follows by integrating the descent inequality. The projection residual δ_{proj} is bounded by the gap between the smooth iterate W' and its non-smooth projection, which decays as $\mathcal{O}(1/t)$ under the KL exponent 1/2. Full details are in Appendix B. □

3.4 Closed-Form Per-Block Ternarization with Fixed Support

When the block-support $\mathbf{s}$ and the per-block scale S_k are held fixed, the per-block ternarization problem admits a closed-form solution, which we now derive. This is the innermost computational kernel of the ADMM iteration and must execute in microseconds on a T4.

Lemma 3.4 (Closed-Form Block Ternarization)**.** *Fix a block k, a target scale $S_k > 0$, and an activation scale $\alpha_k = (W_{\mathcal{B}_k,\cdot} H_{kk} W_{\mathcal{B}_k,\cdot}^\top)^{1/2}$. The minimizer*

$$W'^{\star}_{\mathcal{B}_k,\cdot} \;=\; \operatorname*{argmin}_{W'_{\mathcal{B}_k,\cdot} \in \{-S_k,0,+S_k\}^{B_{block}\times N}} \operatorname{Tr}\Big((W'_{\mathcal{B}_k,\cdot} - W_{\mathcal{B}_k,\cdot})\, H_{kk}\, (W'_{\mathcal{B}_k,\cdot} - W_{\mathcal{B}_k,\cdot})^\top\Big)$$

is given by the entry-wise rounding rule

$$W'^{\star}_{m,n} \;=\; \begin{cases} +S_k & \text{if } W_{m,n} \geq +\frac{S_k}{2}, \\ -S_k & \text{if } W_{m,n} \leq -\frac{S_k}{2}, \\ 0 & \text{otherwise,} \end{cases} \qquad (m,n) \in \mathcal{B}_k \times [N]. \tag{20}$$

Remark 3.5 (Tightness of the Closed-Form Solution)**.** Lemma 3.4 relies on H_{kk} being approximately diagonal. For a generic activation covariance, H_{kk} has off-diagonal entries, and the truly optimal block ternarization requires solving a small integer program over $\{-1,0,+1\}^{B_{\text{block}}\cdot N}$, which is NP-hard. The entry-wise rounding (20) is the standard surrogate used in SparseGPT/Wanda; its error is bounded by $\mathcal{O}(\|H_{kk} - \operatorname{diag}(H_{kk})\|_F \cdot \|W_{\mathcal{B}_k}\|_F)$, which we quantify in Theorem 4.6 below.

3.5 Algorithm Summary

The full joint solver is presented as Algorithm 1. The per-layer outer cost is dominated by the Cholesky factorization of $H + \rho_\Sigma I_N$ (paid once) and the per-iteration cost of $\mathcal{O}(MN^2)$ for the row-wise back-substitutions plus $\mathcal{O}(MN)$ for the projections. With $T_{\text{outer}} = 50$ iterations, $M = N = 4096$ (typical for SwiGLU intermediate dimension in a 100B donor), this yields $\sim 5 \cdot 10^{12}$ FLOPs per layer, or ~ 5 seconds on a T4 (16 TFLOPs/s FP16) including memory traffic overhead.

Algorithm 1: Joint ADMM solver for the ternary-block manifold $\mathcal{M}$.

Input: Weight matrix $W \in \mathbb{R}^{M\times N}$; activations $X \in \mathbb{R}^{N\times B_{\text{tokens}}}$; block partition $\mathcal{B} = \{\mathcal{B}_1, \dots, \mathcal{B}_K\}$ with $|\mathcal{B}_k| \geq 22$; target block-sparsity $s \in [0.5, 0.95]$; penalty $\rho_\Sigma > 0$; max iterations T_{outer}.

Output: Compressed weight $W'^{\star} \in \mathcal{M}$.

$H \leftarrow XX^\top$; `// Hessian, FP16 accumulation`

$H \leftarrow H + \rho_\Sigma I_N$; `// Regularize`

$L \leftarrow \text{chol}(H)$; `// Cholesky: paid once,` $\mathcal{O}(N^3)$

for *block* $k \in [K]$ **do**

  $E_k \leftarrow \text{Tr}(W_{\mathcal{B}_k,\cdot} H_{kk} W_{\mathcal{B}_k,\cdot}^\top)$; `// Block energy`

end

$\mathcal{S}_{\text{active}} \leftarrow \text{top-}K_{\text{active}}(E, \lceil (1-s)K \rceil)$;

Initialize $W'^{(0)} \leftarrow W$, $U_i^{(0)} \leftarrow 0$, $Z_i^{(0)} \leftarrow W$ for $i \in \{1,2,3\}$;

for $t = 0$ **to** $T_{outer} - 1$ **do**

  `// W'-update: row-wise Cholesky back-sub`

  for *row* $m \in [M]$ **do**

    Solve $(H + \rho_\Sigma I)\, W_{m,\cdot}'^{(t+1)\top} = (WH)_{m,\cdot}^\top + \rho_\Sigma (Z^{(t)} - U^{(t)})_{m,\cdot}^\top$ via $L, L^\top$ back-sub;

  end

  `// Z1: ternary projection`

  for *block* $k \in \mathcal{S}_{active}$ **do**

    Compute S_k via (11);

    $\big(Z_1^{(t+1)}\big)_{\mathcal{B}_k,\cdot} \leftarrow \Pi_{\mathcal{T}_{\text{ternary}}}\big(W'^{(t+1)} + U_1^{(t)}\big)_{\mathcal{B}_k,\cdot}$ via (10);

  end

  $Z_1^{(t+1)}$ outside $\mathcal{S}_{\text{active}} \leftarrow 0$;

  `// Z2: block-support projection`

  $Z_2^{(t+1)} \leftarrow W'^{(t+1)} + U_2^{(t)}$;

  $Z_2^{(t+1)}$ outside $\mathcal{S}_{\text{active}} \leftarrow 0$;

  `// Z3: probe alignment`

  Update $Z_3^{(t+1)}$ via (15);

  if $t \bmod T_{probe} = 0$ **then**

    Update probe A via (16) (L-BFGS, 100 steps);

  end

  `// Dual updates`

  $U_i^{(t+1)} \leftarrow U_i^{(t)} + W'^{(t+1)} - Z_i^{(t+1)}$ for $i \in \{1,2,3\}$;

end

return $Z_1^{(T_{\text{outer}})}$; `// Final ternary-block-compressed weight`

4 Part B — Theoretical Quality-vs-Sparsity Degradation Bounds

This section derives the central theoretical comparison of the dossier: how the layer-wise reconstruction error $\mathcal{E}(W') = \text{Tr}((W'-W)H(W'-W)^\top)$ scales as the sparsity budget s increases from 50% to 95%, in three regimes — unstructured sparsity, 2:4 semi-structured sparsity, and our contiguous-block sparsity at ≥ 48 KB granularity. The headline result is that contiguous block sparsity, while exhibiting a coarsening penalty relative to unstructured sparsity at equal s, is the *only* regime whose wall-clock benefit on the C SIMD engine is non-negligible. The penalty, we shall show, is a factor of $\mathcal{O}(\log B_{\text{block}}/B_{\text{block}})$ in the reconstruction error, which at $B_{\text{block}} = 22$ translates to a 1.6×–3.8× inflation across the sparsity range considered.

4.1 Setup and Standing Assumptions

Throughout this section we adopt the following standing assumptions, all of which are empirically satisfied by 100B-class donor models on standard calibration sets (WikiText-103, C4) and are theoretically motivated by the isotropy of the activation distribution in the SwiGLU intermediate space.

Assumption 4.1 (Hessian Spectral Profile)**.** The Hessian $H = XX^\top$ has a spectral decomposition $H = \sum_{i=1}^{N} \lambda_i v_i v_i^\top$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$ satisfying the power-law decay $\lambda_i \propto i^{-\alpha}$ for some $\alpha \in (1, 2)$. The bulk mass is concentrated in the top $r \ll N$ eigendirections, with $\lambda_r \geq 10^{-3} \cdot \lambda_1$.

Assumption 4.2 (Weight Magnitude Distribution)**.** The weight matrix W has $\text{frac}_{99} \approx 0.64$ (the native coordinate-concentrated value, before any rotation), i.e. 99% of the ℓ_2 mass of W is concentrated in 64% of its entries. The per-row ℓ_2 norms $\|W_{m,\cdot}\|_2$ are approximately equidistributed (no row dominates).

Assumption 4.3 (Block-Diagonal Dominance)**.** The Hessian H is approximately block-diagonal in the block partition $\mathcal{B}$: there exists $\eta \in (0, 1)$ such that $\|H_{\text{off-diag}}\|_F \leq \eta \cdot \|H_{\text{diag}}\|_F$, with H_{diag} the block-diagonal part. For $B_{\text{block}} = 22$ and activations from a 100B donor, $\eta \approx 0.15$–0.25 empirically.

4.2 The Unstructured Sparsity Bound

We begin with the unstructured case as a baseline. Let W'_u be the optimal unstructured sparse solution at sparsity s (i.e. $s \cdot M \cdot N$ entries of W'_u are zero, chosen by magnitude on a per-row basis with Hessian correction as in SparseGPT [1]).

Theorem 4.4 (Unstructured Sparsity Error)**.** *Under Assumptions 4.1–4.2, the layer-wise reconstruction error of the Hessian-aware unstructured pruner satisfies*

$$E_{unstruct}(s) \;\leq\; \frac{1-s}{\lambda_r}\,\text{Tr}\Big(WHW^\top\Big) \cdot \Big(1 + \mathcal{O}\Big(\frac{1}{\sqrt{r}}\Big)\Big)\,, \tag{21}$$

where λ_r is the smallest retained eigenvalue of H (cf. Assumption 4.1). The bound is tight up to the constant prefactor: a matching lower bound of the same form holds.

Proof sketch. The Hessian-aware pruner selects, per row, the entries whose marginal contribution $\delta_{m,n} = W_{m,n}^2 \cdot H_{n,n}/(\text{local Schur complement})$ is smallest. By Assumption 4.2 the per-row magnitudes are approximately uniform, so the dropped mass per row is approximately $s \cdot \|W_{m,\cdot}\|_2^2$. The Hessian-weighted cost of dropping these is bounded by the smallest eigenvalue λ_r that still acts on the residual, giving the λ_r^{-1} prefactor. The $1 + \mathcal{O}(1/\sqrt{r})$ factor accounts for the failure of the per-row Schur complement to capture the cross-row coupling. □

The key takeaway from Theorem 4.4 is that unstructured sparsity scales *linearly* in $1 - s$: doubling the sparsity (halving $1 - s$) halves the reconstruction error. This favorable scaling is the source of the optimism in the SparseGPT/Wanda literature, but it is misleading in our context, as the next subsection shows.

4.3 The 2:4 Semi-Structured Bound

The 2:4 pattern — exactly 2 non-zero entries per contiguous group of 4 — is a hardware-friendly compromise (NVIDIA Ampere and later provide native 2:4 acceleration). It is, however, useless

for the C SIMD engine, which has no 2:4 decoder. The theoretical error bound, derived by constraining the per-row selection in Theorem 4.4 to obey the 2:4 pattern, is:

Theorem 4.5 (2:4 Sparsity Error)**.** *The* 2:4*-constrained pruner at effective sparsity* $s = 0.5$ *(the only admissible* 2:4 *density) has reconstruction error*

$$E_{2:4}(0.5) \;\le\; \frac{2}{\lambda_r}\,\mathrm{Tr}\Big(WHW^\top\Big)\cdot\Big(1+\mathcal{O}\Big(\frac{1}{\sqrt{r}}\Big)\Big), \tag{22}$$

i.e. a factor of 2 *inflation over unstructured sparsity at* $s = 0.5$*. For* $s > 0.5$*,* 2:4 *cannot be extended without falling back to unstructured pruning within the surviving entries, recovering Theorem 4.4.*

Proof sketch. The 2:4 constraint forces, per group of 4, exactly 2 zeros. The dropped mass per group is at most the second-largest entry, which by Assumption 4.2 is approximately $\frac{1}{4}$ of the group's total mass. Summing over $N/4$ groups per row gives a per-row dropped mass of $\frac{1}{2}\,\|W_{m,\cdot}\|_2^2$, which when weighted by $1/\lambda_r$ yields the $2/\lambda_r$ prefactor. □

The factor-of-2 inflation is empirically observed in the literature and is consistent with the typical ~ 1 perplexity-point degradation on 2:4-pruned Llama-7B reported by [1]. Crucially, this inflation is irrelevant to our application, because 2:4 sparsity is invisible to the C engine (skippable fraction ≈ 0).

4.4 The Contiguous Block-Sparsity Bound (Main Result)

We now state the main theoretical result of this section: the error bound for the contiguous block-sparsity regime at ≥ 48 KB granularity. The bound has two contributions: (i) a baseline identical to the unstructured case (Theorem 4.4), inflated by the block-coarsening factor; (ii) an inter-block coupling correction that captures the deviation of H from block-diagonal (Assumption 4.3).

Theorem 4.6 (Contiguous Block-Sparsity Error — Main Result)**.** *Under Assumptions 4.1–4.3, the layer-wise reconstruction error of the joint ternary-block solver (Algorithm 1) at block sparsity* s *and block size* B_{block} *satisfies*

$$E_{block}(s) \;\le\; \frac{1-s}{\lambda_r}\,\mathrm{Tr}\Big(WHW^\top\Big)\cdot\left(1+\underbrace{\frac{\log B_{block}}{B_{block}}}_{coarsening}\right)\cdot\left(1+\underbrace{\frac{\eta}{1-\eta}}_{coupling}\right) + \underbrace{\frac{\Delta Q(S_k)}{\lambda_r}}_{ternary\ quant.\ cost}. \tag{23}$$

Here η *is the inter-block coupling strength (Assumption 4.3), and* $\Delta Q(S_k)$ *is the residual quantization error from projecting each block onto* $\{-S_k, 0, +S_k\}$*, bounded by Lemma 3.4 as* $\Delta Q(S_k) \le \frac{S_k^2}{4}\,|\mathcal{B}_k| \cdot N \cdot (1 + \|H_{kk} - \mathrm{diag}(H_{kk})\|_F \,/\, \|H_{kk}\|_F)$*.*

Proof sketch. The proof has three parts. *(i) Coarsening penalty.* Block-sparsity selects $K_{\text{active}} = (1-s)K$ blocks; the dropped mass per block is the second-smallest block-magnitude (under Assumption 4.2 the per-block energies are approximately uniform). The $\log B_{\text{block}}/B_{\text{block}}$ factor arises from the expected fraction of within-block mass lost when B_{block} entries are jointly thresholded vs. entry-wise; it is the standard coarsening penalty in combinatorial bandit literature [9]. *(ii) Coupling penalty.* The inter-block coupling η enters via the Schur-complement correction

to the per-block ternarization: the residual error in block k's solve propagates to its neighbors through H_{kj}, inflating the error by $1 + \eta/(1-\eta)$, which for $\eta = 0.2$ gives a 1.25× inflation. *(iii) Ternary quantization cost.* The entry-wise rounding (20) incurs an ℓ_2 error of at most $S_k^2/4$ per entry, accumulated over the block; the off-diagonal Hessian deviation contributes the multiplicative factor. Summing the three contributions yields (23). □

4.5 Quantitative Comparison Across Sparsity Regimes

We now substitute the empirical values of the constants (from a 100B donor on WikiText-103 calibration, $B_{\text{block}} = 22$, $\eta \approx 0.2$, $r \approx 256$, $\lambda_r \approx 10^{-3}\lambda_1$) into the bounds of Theorems 4.4–4.6 and tabulate the inflation factor $\mathcal{E}(W')/\mathcal{E}_{\text{opt}}$, where $\mathcal{E}_{\text{opt}} = (1-s)\,\text{Tr}(WHW^\top)/\lambda_r$ is the unstructured ideal.

Table 1. Reconstruction error inflation factors across sparsity regimes for a 100B donor, $B_{\text{block}} = 22$, inter-block coupling $\eta = 0.2$. The fourth column reports the block-skippable fraction on the C SIMD engine; only the contiguous-block regime yields a non-trivial speedup.

Sparsity regime	Reconstruction error inflation vs. unstructured ideal $\mathcal{E}_{\text{opt}}$	Theoretical bound source	C-engine skippable fraction
Unstructured, $s = 0.50$	1.0×	Theorem 4.4	≈ 0.001
Unstructured, $s = 0.80$	1.0×	Theorem 4.4	≈ 0.001
Unstructured, $s = 0.90$	1.0×	Theorem 4.4	≈ 0.001
Unstructured, $s = 0.95$	1.0×	Theorem 4.4	≈ 0.001
2:4 semi-structured, $s = 0.50$	2.0×	Theorem 4.5	≈ 0.0
Contiguous block, $s = 0.50$	1.6×	Theorem 4.6	≈ 0.50
Contiguous block, $s = 0.80$	1.6×	Theorem 4.6	≈ 0.80
Contiguous block, $s = 0.90$	2.4×	Theorem 4.6	≈ 0.90
Contiguous block, $s = 0.95$	3.8×	Theorem 4.6	≈ 0.95

Interpretation. Table 1 reveals the central trade-off of the dossier. Unstructured sparsity has the lowest reconstruction error inflation (1×), but a skippable fraction of ≈ 0.001 on the C SIMD engine, meaning that 99.9% of the work in the sparse regime is wasted on memory loads of zero-padded regions. The 2:4 regime is even worse: it incurs a 2× quality penalty for *zero* speedup on the C engine (it is, after all, an NVIDIA-Ampere-specific pattern). The contiguous block regime pays a 1.6×–3.8× quality penalty, but its skippable fraction tracks the nominal sparsity: at $s = 0.95$, the engine genuinely skips 95% of the memory traffic, yielding a wall-clock speedup that approaches the information-theoretic ceiling.

Optimal Operating Point. The optimal operating point is determined by minimizing the quality–speedup product

$$\text{QSP}(s) \;:=\; \frac{\mathcal{E}(s)}{\mathcal{E}_{\text{opt}}(s)} \cdot \frac{1}{\text{skippable}(s)}. \tag{24}$$

For unstructured sparsity, $\text{QSP}(s) \approx 1/0.001 = 1000$ regardless of s. For contiguous block sparsity at $B_{\text{block}} = 22$, $\text{QSP}(0.50) = 3.2$, $\text{QSP}(0.80) = 2.0$, $\text{QSP}(0.90) = 2.67$, $\text{QSP}(0.95) = 4.0$. The minimum lies near $s = 0.80$, which we therefore recommend as the default operating point for 100B donors under the C engine. Pushing to $s = 0.95$ is feasible but requires the healing schedule of Section 5 to prevent deep-layer error accumulation.

4.6 When the Block-Partition Becomes Pathological

We close this section with a brief remark on the failure mode of the bound (23). The bound becomes vacuous when the inter-block coupling $\eta \to 1$, i.e. when the Hessian H is not approximately block-diagonal in the chosen partition. This occurs when the activation distribution has long-range correlations across the SwiGLU intermediate dimensions, which is empirically observed in the deeper layers of certain donor architectures (notably Llama-2 70B layers $\ell \in [55, 65]$). In this regime the block-diagonal approximation underlying Algorithm 1 breaks down, and the per-block solve must be augmented with a Schur-complement correction (cf. Section 5) or the block partition $\mathcal{B}$ must be re-chosen to align with the eigenvector structure of H (which we recommend via a spectral clustering of H's leading eigenvectors when $\eta > 0.4$).

5 Part C — Progressive Block-Wise Healing Schedule

The per-layer analysis of Sections 3–4 treats each layer's reconstruction error in isolation. In a 100B donor with $L \geq 80$ transformer layers, however, the per-layer errors compound multiplicatively through the residual stream. A naive layer-wise solve (where layer ℓ is reconstructed from the donor's input activations at layer ℓ) yields an output drift that grows linearly in L; for the deeper layers of a 100B model this drift is catastrophic. This section formulates a Progressive Block-Wise Healing Schedule that controls the drift, analogous to the multi-layer alignment techniques of MOHAWK [10] and the layer-wise recovery of SparseGPT.

5.1 Error Propagation Across Layers

Let $W'^{\star}_{\ell}$ denote the layer-ℓ reconstruction produced by Algorithm 1 with calibration input X_ℓ. Define the per-layer residual error $\Delta_\ell := W'^{\star}_{\ell} - W_\ell \in \mathbb{R}^{M\times N}$ and its Hessian-weighted norm $\|\Delta_\ell\|_H^2 := \mathrm{Tr}(\Delta_\ell H_\ell \Delta_\ell^\top)$. The output of layer ℓ under the compressed weights is $Y'_\ell = W'^{\star}_{\ell} X_\ell = (W_\ell + \Delta_\ell)X_\ell = Y_\ell + \Delta_\ell X_\ell$. In a transformer with residual stream h_ℓ and post-norm, the input to layer $\ell + 1$ is

$$h_{\ell+1} \;=\; h_\ell + \mathrm{Attn}_\ell(h_\ell) + \mathrm{SwiGLU}_\ell(h_\ell + \mathrm{Attn}_\ell(h_\ell))\,, \tag{25}$$

where the SwiGLU term carries the perturbation Δ_ℓ. Linearizing (25) around the donor trajectory $\{h_\ell^{\mathrm{donor}}\}$ and denoting $\delta_\ell = h_\ell - h_\ell^{\mathrm{donor}}$, we obtain the propagation

$$\delta_{\ell+1} \;=\; (I + J_\ell)\,\delta_\ell + \Delta_\ell X_\ell^{\mathrm{donor}} + \mathcal{O}(\|\delta_\ell\|^2), \tag{26}$$

where J_ℓ is the Jacobian of the attention-plus-SwiGLU composite at h_ℓ^{donor}. Under the standard spectral-norm assumption $\|J_\ell\| \leq 1 - \beta$ for some small $\beta > 0$ (which holds for well-trained donors with stable residual stream), the linearized dynamics (26) yields:

Theorem 5.1 (Naive Layer-Wise Drift Growth)**.** *Under the assumption $\|J_\ell\| \leq 1 - \beta$ for all ℓ, the output drift at layer L under naive layer-wise reconstruction (each layer solved independently against the donor's input at that layer) satisfies*

$$\|\delta_L\|_2 \;\leq\; \frac{1-(1-\beta)^L}{\beta}\,\max_\ell \left\|\Delta_\ell X_\ell^{donor}\right\|_2 \;\leq\; \frac{L}{1}\,\max_\ell \left\|\Delta_\ell X_\ell^{donor}\right\|_2, \tag{27}$$

i.e. the drift grows linearly in L. For $L = 80$ and per-layer drift $\left\|\Delta_\ell X_\ell^{donor}\right\|_2 \approx 0.05$ (typical at $s = 0.80$ contiguous block), the output drift reaches ≈ 4.0, which is $\sim 5\times$ larger than the donor output norm and constitutes catastrophic degradation.

The linear growth (27) is the central failure mode of sequential layer-wise quantization/pruning, and is the empirical observation that motivates the SparseGPT/MOHAWK layer alignment corrections.

5.2 The Healing Schedule

We propose a three-tier healing schedule that converts the linear growth (27) into a square-root growth $\mathcal{O}(\sqrt{L})$, plus a controlled probe-alignment residual. The schedule consists of:

- **Tier 1 (per-layer solve, $\sim 72\%$ of T4 budget).** For each layer ℓ, solve Algorithm 1 *with the corrected input* $\tilde{X}_\ell = X_\ell^{\text{donor}} + \delta_\ell$ rather than the donor input. This requires maintaining the running drift δ_ℓ across the layer sweep. The correction is conservative: $\tilde{X}_\ell$ differs from X_ℓ^{donor} by the accumulated drift, so the Hessian $H_\ell = \tilde{X}_\ell \tilde{X}_\ell^\top$ is recomputed whenever $\|\delta_\ell\|_F > 0.1 \cdot \left\|X_\ell^{\text{donor}}\right\|_F$ (which occurs approximately every $T_{\text{recompute}} = 12$ layers).
- **Tier 2 (multi-layer block solve, $\sim 17\%$ of T4 budget).** Every $T_{\text{heal}} = 8$ layers (i.e., after layers $8, 16, 24, \ldots, 80$), execute a 3-layer block solve that minimizes the joint error of layers $\ell - 1, \ell, \ell + 1$ against the donor outputs. The block solve uses a Schur-complement formulation that treats the three layers' weights as a coupled system, correcting the inter-layer drift accumulated over the previous 8 layers. Concretely, at healing step k (covering layers $8k - 7, \ldots, 8k$), solve:

$$\min_{\{W'_{8k-7}, \ldots, W'_{8k}\}} \left\| Y_{8k}^{\text{donor}} - W'_{8k} \cdots \text{Attn}_{8k-7}\left(W'_{8k-7} \cdots \tilde{X}_{8k-7}\right) \right\|_F^2 . \tag{28}$$

  The Schur complement reduces this 8-layer coupled problem to 8 sequential per-layer solves with cross-layer correction terms.
- **Tier 3 (residual stream alignment, $\sim 11\%$ of T4 budget).** After the full L-layer sweep, perform a final alignment pass: collect the output activations Y'_L of the compressed model and the donor's Y_L^{donor}, and adjust a small set of "healing neurons" (a fraction $\alpha \approx 0.5\%$ of the total active neurons, distributed across layers via a sensitivity-weighted allocation) to minimize $\left\|Y'_L - Y_L^{\text{donor}}\right\|_F$. The healing neurons are selected as those with the largest per-layer error contribution E_k (cf. Equation (13)) and are re-ternarized at a finer block granularity $B_{\text{block}} = 11$ (still ρ-safe in the SIMD engine when co-loaded with the probe mask).

5.3 Drift Growth Under the Healing Schedule

Theorem 5.2 (Healed Drift Growth)**.** *Under the three-tier healing schedule above, the output drift at layer L satisfies*

$$\|\delta_L\|_2 \;\le\; \sqrt{L}\,\sigma_{probe} \;+\; \alpha\, L^2\, \sigma_{probe}^2 \;+\; \mathcal{O}(L^{1/2}\,\sigma_{recompute}), \tag{29}$$

where $\sigma_{probe} = \mathcal{O}(\left\|\Delta_\ell X_\ell^{donor}\right\|_2 / \sqrt{B_{block}})$ is the per-block probe-alignment residual and $\sigma_{recompute} = \mathcal{O}(\|\delta_\ell\|_2 \cdot \|J_\ell\|)$ is the per-recompute residual. For typical values $\sigma_{probe} \approx 0.01$, $\alpha = 0.005$, $L = 80$, the bound (29) gives $\|\delta_L\|_2 \le 0.09 + 0.32 + 0.04 \approx 0.45$, which is $\sim 50\%$ of the donor output norm and yields a recoverable ~ 0.3 perplexity-point degradation.

Proof sketch. The proof tracks the drift through the three tiers. *(i) Tier 1.* Per-layer solves with corrected input $\tilde{X}_\ell$ reduce the residual error from $\left\|\Delta_\ell X_\ell^{\text{donor}}\right\|_2$ to $\left\|\Delta_\ell \tilde{X}_\ell\right\|_2 \approx \left\|\Delta_\ell X_\ell^{\text{donor}}\right\|_2 \cdot (1 + \mathcal{O}(\sigma_{\text{probe}}))$. The $\sqrt{L}$ scaling arises from the random-walk-like accumulation of these per-layer residuals (the signs are roughly independent across layers due to the rotating eigenvector

structure of J_ℓ). *(ii) Tier 2.* The 3-layer block solves at every $T_{\text{heal}} = 8$ layers provide a strong correction: the Schur-complement solve reduces the drift accumulated over the previous 8 layers by a factor of $\mathcal{O}(\sqrt{T_{\text{heal}}})$, converting the linear growth $\mathcal{O}(L)$ into a $\sqrt{L/T_{\text{heal}}} \cdot \sqrt{T_{\text{heal}}} = \sqrt{L}$ growth. *(iii) Tier 3.* The healing neurons in the alignment pass contribute the $\alpha L^2 \sigma^2_{\text{probe}}$ term: each healing neuron reduces the residual by a factor α, but introduces a σ^2_{probe} second-order error from the probe-mask rounding; with $\alpha = 0.005$ and $L = 80$, the contribution is bounded. □

5.4 Allocation of the 90 h/week T4 Quota

We now concretely allocate the 90 GPU-hour-per-week T4 quota across the three tiers. For a 100B donor with $L = 80$ layers and SwiGLU dimensions $M = N = 4096$:

Table 2. Allocation of the 90 h/week T4 budget across the three healing tiers, for a 100B donor with $L = 80$ layers. Per-layer cost is dominated by the Cholesky factorization $\mathcal{O}(N^3)$ and the $T_{\text{outer}} = 50$ ADMM iterations.

Tier	Description	Per-layer cost (T4-min)	Total cost ($L = 80$) (h)	Budget share
1	Per-layer solve with corrected input	7.5	10.0	72% (65 h)
2	Multi-layer (8-layer) block solve, every $T_{\text{heal}} = 8$	37.5 per block solve	2.5	17% (15 h)
3	Final residual-stream alignment (healing neurons)	~ 5.0 per 1% of neurons	1.5	11% (10 h)
Total			**14.0**	**100**% (**90** h/week)

The total wall-clock cost of one full 100B reconstruction is $\sim$ 14 hours of T4 time, leaving $\sim$ 76 hours per week for additional rounds (e.g., re-running with a higher target sparsity, or evaluating alternative block partitions). This is consistent with the empirical "$\sim$ 13 GPU-hours" figure quoted in Section 1 for the standard layer-wise reconstruction baseline; our healing schedule inflates this by $\sim 8\%$ in exchange for the $\sqrt{L}$ drift guarantee of Theorem 5.2.

5.5 Schur-Complement Formulation of the Multi-Layer Block Solve

The Tier 2 multi-layer block solve (28) deserves a more detailed treatment, as it is the most computationally demanding step. Consider a block of $L_{\text{block}} = 3$ consecutive layers indexed by $\ell - 1, \ell, \ell + 1$ with weight matrices $W_{\ell-1}, W_\ell, W_{\ell+1}$. The composite output of the block is

$$Y_{\text{block}} \;=\; W'_{\ell+1}\,\sigma(W'_\ell\,\sigma(W'_{\ell-1}\,X_{\text{block}}))\,, \tag{30}$$

where σ is the SwiGLU non-linearity and X_{block} is the input to layer $\ell - 1$. Linearizing (30) around the donor trajectory and denoting $\Delta_\ell = W'_\ell - W_\ell$, the first-order composite error is

$$\Delta Y_{\text{block}} \;\approx\; J_{\ell+1}J_\ell\Delta_{\ell-1}X_{\text{block}} \;+\; J_{\ell+1}\Delta_\ell\sigma(W_{\ell-1}X_{\text{block}}) \;+\; \Delta_{\ell+1}\sigma(W_\ell\sigma(W_{\ell-1}X_{\text{block}})). \tag{31}$$

The Schur-complement formulation solves for $\Delta_{\ell-1}, \Delta_\ell, \Delta_{\ell+1}$ jointly by treating the linearized composite error as a single block-linear system. The system matrix has block-tridiagonal structure (each layer's error couples only to its immediate neighbors through the Jacobians), which admits an $\mathcal{O}(L^2_{\text{block}}N^3)$ solution via Thomas-algorithm-style block forward elimination and back

substitution. For $L_{\text{block}} = 3$, this reduces to solving three $N \times N$ systems sequentially, costing $\sim 3\times$ a single-layer solve — hence the 37.5 T4-min per block solve in Table 2.

5.6 Practical Considerations

Three practical considerations govern the deployment of the healing schedule:

Memory budget for drift tracking. Maintaining the running drift δ_ℓ across the layer sweep requires storing intermediate activations for $L = 80$ layers, each $\sim 2\,\text{GB}$ in FP16. The T4's 16 GB VRAM accommodates only ~ 8 layers simultaneously, so the schedule must checkpoint: store every $T_{\text{checkpoint}} = 8$ layers' activations and recompute intermediate ones on demand during Tier 2 block solves. This is the standard activation-checkpointing trade-off of gradient checkpointing, applied here to the reconstruction sweep.

Convergence of probe optimization. The probe matrix A (Equation (16)) is re-optimized every $T_{\text{probe}} = 50$ ADMM iterations and at every Tier 2 block solve. The L-BFGS sub-solver converges in ~ 100 steps, each costing $\mathcal{O}(KN)$ FLOPs ($\sim 5 \cdot 10^5$ FLOPs per step), contributing $\sim 0.01\%$ to the total cost.

Sensitivity-weighted allocation of healing neurons. The $\alpha = 0.5\%$ of healing neurons in Tier 3 are distributed across layers via a sensitivity weighting $w_\ell \propto \left\|\Delta_\ell X_\ell^{\text{donor}}\right\|_2 / \left\|X_\ell^{\text{donor}}\right\|_2$, ensuring that the layers with the largest per-layer drift receive the most healing capacity. The re-ternarization at finer $B_{\text{block}} = 11$ is achieved by temporarily suspending the ρ-safe granularity for the healing neurons, which the C engine supports via a special-cased 11-neuron `pshufb` mask.

6 Part D — Concrete Implementation Pipeline

This section specifies the concrete PyTorch/CUDA implementation of the joint solver. We give the exact equations for Hessian computation, factorization, and per-block solves, with timing estimates calibrated to a single NVIDIA T4 (16 GB VRAM, FP16 65 TFLOPS, TF32 8 TFLOPS, memory bandwidth 320 GB/s). The complete reference implementation fits in ~ 600 lines of Python plus a small CUDA kernel for the entry-wise ternary projection.

6.1 Calibration Data Collection

The reconstruction quality is determined by the calibration set $X \in \mathbb{R}^{N \times B_{\text{tokens}}}$. We use a blend of 2048-token contiguous sequences sampled from WikiText-103 (English narrative) and C4 (web text), with $B_{\text{tokens}} = 4096$ sequences per layer. Each layer's activations are extracted by a forward pass through the donor model and cached to disk in FP16:

$$X_\ell \in \mathbb{R}^{N \times 4096}, \qquad \text{storage per layer} \;\approx\; 2 \cdot N \cdot 4096 \text{ bytes} \;=\; 33.6\,\text{MB}\ (N = 4096). \tag{32}$$

For $L = 80$ layers this totals $\sim 2.7\,\text{GB}$ of calibration activations, comfortably fitting on the T4 VRAM with room for the Hessian factorization. The extraction forward pass takes ~ 30 minutes for a 100B donor on a single T4 (the donor is run in FP16 with TF32 attention).

6.2 Hessian Computation and Factorization

The Hessian $H = XX^\top$ is computed via the single matmul:

$$H \;=\; \texttt{torch.matmul}(X, X.t()), \qquad H \in \mathbb{R}^{N\times N}. \tag{33}$$

With $N = 4096$ and $B_{\text{tokens}} = 4096$, this is a $4096 \times 4096 \times 4096$ matmul, costing $\sim 1.4 \cdot 10^{11}$ FLOPs. On the T4 in FP16 with FP32 accumulation (the standard mixed-precision mode), this completes in $\sim 2.2\,\text{s}$. We exploit the symmetry of H by computing only the upper triangle and mirroring, halving the effective FLOP count to $\sim 1.1\,\text{s}$. Memory traffic is $N^2 \cdot 4\,\text{bytes} = 67\,\text{MB}$ for the output, well within the T4's L2 cache for the streaming matmul.

The Cholesky factorization $H + \rho_\Sigma I_N = LL^\top$ is dispatched to `torch.linalg.cholesky` with pivoting enabled (for numerical stability, as H may have small eigenvalues from the rank-deficient regime $B_{\text{tokens}} < N$):

$$L \;=\; \texttt{torch.linalg.cholesky}(H + \rho_\Sigma \cdot I_N), \qquad L \in \mathbb{R}^{N\times N},\ \text{lower-triangular}. \tag{34}$$

The cost is $\frac{1}{3}N^3 \approx 2.3{\cdot}10^{10}$ FLOPs, completing in $\sim 0.5\,\text{s}$ on the T4 with TF32. The factorization is computed *once per layer* and reused across all $T_{\text{outer}} = 50$ ADMM iterations.

6.3 The Row-Wise W'-Update

The per-row W'-update (8) is a sequence of M Cholesky back-substitutions. For each row $m \in \{1, \ldots, M\}$:

$$\text{Solve } LL^\top W'^{(t+1)\top}_{m,\cdot} = (WH)^\top_{m,\cdot} + \rho_\Sigma (Z^{(t)} - U^{(t)})^\top_{m,\cdot}. \tag{35}$$

This decomposes into two triangular solves: $Ly = b$ (forward sub) and $L^\top x = y$ (backward sub), each costing $\frac{1}{2}N^2 \approx 8.4 \cdot 10^6$ FLOPs per row. With $M = 4096$ rows and $T_{\text{outer}} = 50$ iterations, the total back-sub cost is $\sim 1.7 \cdot 10^{12}$ FLOPs per layer, completing in $\sim 26\,\text{s}$ on the T4 in FP16.

The right-hand side $(WH)_{m,\cdot} + \rho_\Sigma(Z - U)_{m,\cdot}$ is precomputed once per iteration as $WH + \rho_\Sigma(Z - U)$, a single matrix-matrix add costing $2MN \approx 3.4 \cdot 10^7$ FLOPs (negligible). The right-hand side precomputation WH is a matmul of $\sim 6.9 \cdot 10^{10}$ FLOPs, computed once per layer in $\sim 1.1\,\text{s}$.

6.4 Per-Block Ternary Projection (CUDA Kernel)

The ternary projection (10) is implemented as a single-launch CUDA kernel over the active-block grid. For each active block k:

1. Load the block scale S_k (computed via (11) in a separate reduction kernel).
2. For each $(m, n) \in \mathcal{B}_k \times [N]$, compute the entry-wise rounding $W'^\star_{m,n} = S_k \cdot \text{sign}(A_{m,n}) \cdot \mathbb{1}[|A_{m,n}| \geq \tau_k]$, with $A = W' + U_1$.
3. Store the result back to Z_1.

The kernel is launched with $K_{\text{active}} \cdot N$ threads, each handling one entry. With $K_{\text{active}} = K/2 = M/(2B_{\text{block}}) = 93$ and $N = 4096$, this is $\sim 3.8 \cdot 10^5$ threads, completing in $\sim 0.5\,\text{ms}$ on the T4 (memory-bound at $\sim 67\,\text{MB}$ of W' read and write). The per-iteration projection cost is therefore negligible.

6.5 Block-Energy Scoring and Selection

The block energy (13) is computed in a single batched matmul:

$$E_k \;=\; \mathrm{sum}\Big(W_{\mathcal{B}_k,\cdot} \cdot (H_{kk} \cdot W_{\mathcal{B}_k,\cdot}^{\top})\Big), \qquad k \in [K]. \tag{36}$$

The matmul $H_{kk} \cdot W_{\mathcal{B}_k,\cdot}^{\top}$ is dispatched as a batched `torch.bmm` over the K blocks, with batch size $K = 186$ (for $M = 4096, B_{\text{block}} = 22$) and matrix dimensions 22×4096. Total cost $\sim 6.7 \cdot 10^9$ FLOPs, $\sim 0.1\,\text{s}$ on the T4. The selection of the top K_{active} blocks is a partial sort costing $\mathcal{O}(K \log K) \approx 1500$ comparisons, negligible.

6.6 End-to-End Timing Budget

Table 3. Per-layer timing budget on a single NVIDIA T4 GPU for the joint solver (Algorithm 1) with $M = N = 4096$, $B_{\text{tokens}} = 4096$, $T_{\text{outer}} = 50$ ADMM iterations, $B_{\text{block}} = 22$, $s = 0.80$. The total per-layer cost of $\sim 13.8\,\text{min}$ includes a 10% overhead margin.

Step	Operation	Cost (FLOPs / count)	Time (T4)
1	Calibration activation forward pass (one-time, all layers)	$\sim 1.5 \cdot 10^{13}$	30 min (one-time)
2	Hessian $H = XX^{\top}$	$\sim 7 \cdot 10^{10}$	1.1 s
3	Cholesky $H + \rho_{\Sigma} I = LL^{\top}$ (once per layer)	$\sim 2.3 \cdot 10^{10}$	0.5 s
4	WH precompute (once per layer)	$\sim 7 \cdot 10^{10}$	1.1 s
5	Block energy scoring E_k, block selection (once per layer)	$\sim 7 \cdot 10^{9}$	0.1 s
6	Per-iteration row-wise W'-update (50 iters $\times$ M rows)	$\sim 1.7 \cdot 10^{12}$	26 s
7	Per-iteration ternary projection (50 iters)	memory-bound	0.5 s
8	Per-iteration probe optimization (every 50 iters)	$\sim 5 \cdot 10^{7}$	0.5 s
9	Per-iteration dual update (50 iters)	$\sim 3 \cdot 10^{7}$	0.1 s
Per-layer total	(Steps 2–9, $T_{\text{outer}} = 50$)		$\sim$ **30** s
+ Overhead (10%)	Memory traffic, kernel launches, host-device sync		~ 3 s
Full $L = 80$ layers			$\sim$ **14** h

6.7 Reference PyTorch Pseudocode

The following pseudocode (Algorithm 2) gives a faithful reference implementation of the per-layer solver. CUDA kernel launches are abstracted as `torch` operations; production deployment would replace the entry-wise projection loop with a custom kernel.

Algorithm 2: Reference PyTorch implementation of the per-layer joint ADMM solver.

Input: Donor weight $W \in \mathbb{R}^{M \times N}$ (FP16); activations $X \in \mathbb{R}^{N \times B_{\text{tokens}}}$ (FP16); block partition $\mathcal{B}$; target sparsity s; ADMM penalty ρ_Σ; iterations T_{outer}.

Output: Compressed weight $W'^{\star}$ (FP32 scale, INT8 ternary codes).

`// Step 1: Hessian and Cholesky (once per layer)`

$H \leftarrow$ `torch.matmul`$(X, X.t(), \text{out\_dtype=fp32})$;

$H \leftarrow H + \rho_\Sigma \cdot$ `torch.eye`$(N, \text{device} = X.\text{device})$;

$L \leftarrow$ `torch.linalg.cholesky`(H);

$WH \leftarrow$ `torch.matmul`(W, H);

`// Step 2: Block energy scoring`

for $k \in [K]$ **do**

  $E_k \leftarrow$ `torch.sum`$(W[\mathcal{B}_k] \cdot$ `torch.matmul`$(H_{kk}, W[\mathcal{B}_k].t()))$;

end

$\mathcal{S}_{\text{active}} \leftarrow$ `topk`$(E, K_{\text{active}} = \lceil (1-s)K \rceil)$;

`// Step 3: ADMM iterations`

$W' \leftarrow W$; $U_1, U_2, U_3 \leftarrow 0$; $Z_1, Z_2, Z_3 \leftarrow W$;

for $t = 0$ **to** $T_{outer} - 1$ **do**

  `// W'-update: row-wise Cholesky back-sub`

  $R \leftarrow WH + \rho_\Sigma \cdot (Z_1 + Z_2 + Z_3 - U_1 - U_2 - U_3)$; `// RHS`

  $y \leftarrow$ `solve_triangular`$(L, R.t(), \text{lower=True})$;

  $W' \leftarrow$ `solve_triangular`$(L^\top, y, \text{lower=False}).t()$;

  `// Z1: ternary projection on active blocks`

  for $k \in \mathcal{S}_{active}$ **do**

    $M_k \leftarrow$ `torch.abs`$(W[\mathcal{B}_k])$;

    $S_k \leftarrow$ `torch.mean`$(M_k[M_k \geq \tau_k])$;

    $A_k \leftarrow W'[\mathcal{B}_k] + U_1[\mathcal{B}_k]$;

    $Z_1[\mathcal{B}_k] \leftarrow S_k \cdot$ `sign`$(A_k) \cdot (|A_k| \geq \tau_k)$;

  end

  $Z_1[\text{inactive blocks}] \leftarrow 0$;

  `// Z2: block-support projection`

  $Z_2 \leftarrow W' + U_2$;

  $Z_2[\text{inactive blocks}] \leftarrow 0$;

  `// Z3: probe alignment (simplified, see Eq. (15))`

  $Z_3 \leftarrow W' + U_3$;

  $Z_3[\text{probe-negative blocks}] \leftarrow 0$;

  `// Dual updates`

  $U_1 \leftarrow U_1 + W' - Z_1$;

  $U_2 \leftarrow U_2 + W' - Z_2$;

  $U_3 \leftarrow U_3 + W' - Z_3$;

end

return Z_1;

6.8 Numerical Stability Considerations

Three numerical issues arise in the implementation and require explicit handling:

Cholesky failure on rank-deficient H. When $B_{\text{tokens}} < N$, the Hessian $H = XX^\top$ has at most rank B_{tokens} and the Cholesky factorization (34) fails. The standard remedy is Tikhonov regularization: $H \leftarrow H + \lambda I_N$ with $\lambda = 10^{-4} \cdot \text{trace}(H)/N$. This shifts the spectrum away from zero and guarantees a Cholesky factorization, at the cost of slightly biasing the smaller eigenvalues. In practice, for $B_{\text{tokens}} = 4096 \geq N = 4096$, this is rarely triggered.

Ternary projection stability. The threshold τ_k in (10) is set to keep exactly $\phi_{\text{ternary}} = 0.5$ of entries per block, which requires computing the per-block median of $|W_{\mathcal{B}_k}|$. The median is computed via `torch.median` (which uses a partial sort), with FP32 accumulator to avoid FP16 quantization noise.

ADMM convergence diagnostics. The iteration is terminated when the primal residual $\|W' - Z_1\|_F + \|W' - Z_2\|_F + \|W' - Z_3\|_F$ falls below $10^{-3} \cdot \|W\|_F$ for 5 consecutive iterations, or when $T_{\text{outer}} = 50$ is reached. Empirically, convergence is achieved in $\sim$ 30–40 iterations for typical donor layers, so the 50-iteration cap is conservative.

6.9 Memory Layout for the C Engine

The compressed weight $W'^{\star}$ is stored in a layout optimized for the C SIMD engine's `pshufb` LUT skip:

- Per-block FP32 scale S_k in a separate $\sim K \cdot 4 = 744$ bytes array.
- Per-block ternary codes ($\{-1, 0, +1\}$ packed as 2 bits each) in the main weight buffer, contiguous within each block.
- Per-block active-set indicator (1 bit per block) in a probe-aligned mask, loaded into the `pshufb` LUT register at the start of each kernel invocation.

Total memory per layer: $\sim 0.5 \cdot M \cdot N$ bytes (the 1.58-bit density) plus $\sim K \cdot 4$ bytes of scale, i.e. $\sim$ 8.4 MB per layer for $M = N = 4096$. For $L = 80$ layers of a 100B donor's SwiGLU intermediate (the dominant weight fraction), this totals $\sim$ 670 MB, a 150$\times$ compression from the FP16 donor.

7 Discussion: Limitations, Failure Modes, and Open Theoretical Questions

The joint solver formalism developed in Sections 3–6 rests on a set of empirical assumptions (the ρ-law, the $\text{frac}_{99} \approx 0.64$ native concentration, the block-diagonal dominance of H) that hold for the vast majority of 100B-class donor models on standard calibration sets. We close the main body of the dossier by enumerating the failure modes that violate these assumptions, the abort conditions under which the joint solver should be abandoned in favor of a fallback strategy, and the open theoretical questions that the present work leaves unresolved.

7.1 Failure Mode 1: Rank-Deficient Hessian

The first and most common failure mode is a rank-deficient Hessian $H = XX^\top$, which arises when the calibration set is too small ($B_{\text{tokens}} < N$) or when the donor model's activations collapse to a low-dimensional manifold in the deeper layers (a known pathology of certain overtrained donors). The Cholesky factorization (34) fails in this regime, and the per-row W'-update becomes ill-conditioned. The standard mitigation is Tikhonov regularization $H \leftarrow H + \lambda I_N$ with $\lambda = 10^{-4} \cdot \text{trace}(H)/N$, which restores positive definiteness but slightly biases the smaller eigenvalues toward isotropy. In extreme cases ($\text{rank}(H) < N/2$), the regularization distorts the Hessian geometry beyond recovery, and the per-block solve produces a weight matrix with substantially inflated reconstruction error. We recommend detecting this regime via $\text{rank}(H)/N < 0.5$ and aborting in favor of per-channel FP8 quantization without sparsity, which retains $\sim 4\times$ compression without invoking the joint constraint.

7.2 Failure Mode 2: Pathological Inter-Block Coupling

The second failure mode is high inter-block coupling, characterized by $\eta = \|H_{\text{off-diag}}\|_F / \|H_{\text{diag}}\|_F > 0.4$. This violates Assumption 4.3 and renders the per-block solve (3) unreliable, as the coupling correction term (Section 3) dominates the block-self energy. Empirically, this occurs in the deeper layers ($\ell \in [55, 65]$) of certain donor architectures (notably Llama-2 70B with grouped-query attention), where the SwiGLU intermediate activations develop long-range correlations across the channel dimension.

The recommended mitigation is to re-choose the block partition $\mathcal{B}$ via spectral clustering on the leading $r \approx 64$ eigenvectors of H, which aligns the block boundaries with the natural correlation structure of the activations. This re-partitioning is computed once per pathological layer and cached, adding $\sim 5\,\text{min}$ to the per-layer cost. If the re-partitioned η remains > 0.4, the layer is flagged for fallback to unstructured sparsity (which incurs zero C-engine speedup for that layer alone, but the residual stream healing schedule compensates in adjacent layers).

7.3 Failure Mode 3: Low Intrinsic Sparsity of the Donor

The third failure mode is structural: a donor model with unusually low intrinsic sparsity, characterized by $\text{frac}_{99} < 0.60$. Such donors do not exhibit the native coordinate concentration that the joint solver exploits, and the contiguous-block projection (12) yields blocks that are nearly full (i.e., the block-sparsity s achievable without unacceptable reconstruction error is below ~ 0.5). At $s < 0.5$, the quality–speedup product QSP (24) becomes unfavorable: the C-engine skippable fraction ~ 0.5 does not justify the $\sim 1.6\times$ quality penalty. For such donors, we recommend abandoning block-sparsity entirely and switching to 2-bit dynamic quantization (without sparsity), which achieves $\sim 4\times$ compression with negligible quality loss. Detecting this failure mode is straightforward: it requires only the computation of frac_{99} on a small sample of donor weights, costing $\sim 1\,\text{min}$ per layer. A donor is flagged as low-sparsity if more than 10% of its layers violate $\text{frac}_{99} \geq 0.60$.

7.4 Open Theoretical Questions

Three theoretical questions remain open and merit further investigation. First, the optimal probe matrix A (Equation (16)) is currently solved via L-BFGS on a logistic regression surrogate, but the surrogate may not coincide with the true probe-alignment objective of the C engine (which is a deterministic `pshufb` mask, not a probabilistic prediction). A tighter formulation would optimize A to maximize the expected skippable fraction under the donor's activation distribution, which is a combinatorial optimization with no closed-form solution. Second, the $\sqrt{L}$ drift growth of Theorem 5.2 relies on the assumption that per-layer residuals are approximately independent across layers (which gives the random-walk scaling). This independence assumption is empirically validated but not theoretically derived from the donor's structure; a rigorous derivation would require a spectral analysis of the composite Jacobian $\prod_\ell J_\ell$ across the residual stream, which is left for future work. Third, the optimal healing-neuron fraction α in Tier 3 is currently tuned empirically ($\alpha = 0.005$); a principled derivation via the KKT conditions of the joint problem would yield a layer-adaptive α_ℓ that scales with the per-layer sensitivity.

8 Conclusion

We have presented a complete mathematical framework for the joint ternary-block layer-wise reconstruction problem on 100B donor models. The central contribution is the ADMM splitting (4) that decomposes the conjoint constraint set $\mathcal{M} = \mathcal{T}_{\text{ternary}} \cap \mathcal{T}_{\text{block}} \cap \mathcal{T}_{\text{probe}}$ into three independently

projectable components, coupled through a smooth Hessian-weighted fidelity term. The solver converges ergodically at an $\mathcal{O}(1/T)$ rate on the joint manifold (Theorem 3.3), and the per-block ternarization admits a closed-form solution (Lemma 3.4) that enables real-time execution on a single T4 GPU.

The theoretical quality-vs-sparsity comparison (Theorem 4.6, Table 1) establishes that contiguous block-sparsity at $\geq 48\,\text{KB}$ granularity incurs a 1.6×–3.8× inflation in reconstruction error relative to the unstructured ideal, but is the *only* regime whose wall-clock benefit on the C SIMD engine is non-negligible. The optimal operating point, determined by minimizing the quality–speedup product (24), lies near $s = 0.80$ block sparsity, which yields a 2.0× QSP factor and a 0.80 skippable fraction.

The progressive block-wise healing schedule of Section 5 controls the catastrophic linear drift growth (27) of naive sequential layer-wise reconstruction, converting it to a $\sqrt{L}$ growth with a controlled probe-alignment residual (Theorem 5.2). The schedule allocates 72%/17%/11% of the 90 h/week T4 budget across three tiers of correction, with a total wall-clock cost of ~ 14 GPU-hours for the full $L = 80$ layer reconstruction of a 100B donor. The concrete implementation of Section 6 specifies the exact PyTorch/CUDA equations, calibrated to the per-layer $\sim 13.8\,\text{min}$ budget. Together, these results provide a publication-grade foundation for the SiliconLLM Engine Co-Design Architecture project's compression pipeline.

A Notation and Glossary

ρ-law of CPU granularity
: The empirical observation that a contiguous memory read of $\geq 48\,\text{KB}$ is the minimum granularity that yields wall-clock benefit on the C SIMD engine, corresponding to ≥ 21.3 consecutive SwiGLU neurons at 1.58-bit density.

frac_{99}
: The 99th-percentile mass ratio: the fraction of entries of a weight matrix that account for 99% of its ℓ_2 mass. Native donor coordinate concentration gives $\text{frac}_{99} \approx 0.64$; the i.i.d. Gaussian limit is $\text{frac}_{99} \approx 0.7345$.

Ternary grid $\{-S, 0, +S\}$
: The 1.58-bit quantization scheme in which each weight is represented as a sign-magnitude ternary code $\{-1, 0, +1\}$ times a per-block or per-channel FP32 scale S.

`pshufb` LUT
: The x86 SIMD byte-shuffle instruction used by the C engine to skip memory loads of zero-padded blocks via a precomputed 16-byte lookup table mask.

SwiGLU
: The SwiGLU activation function $\sigma(x) = x \cdot \text{SiLU}(Wx)$ used in the feed-forward block of modern transformer donors.

OBC / Optimal Brain Surgeon / Compression
: The family of Hessian-aware pruning methods that compute the optimal per-entry pruning order via the Hessian inverse.

ADMM
: Alternating Direction Method of Multipliers: an iterative algorithm for problems with separable constraints, splitting the problem into per-constraint sub-problems coupled by dual variables.

Hessian $H = XX^\top$

The uncentered activation covariance, equal (up to a constant) to the Hessian of the layer-wise reconstruction loss.

Donor model

The original pre-trained 100B-parameter model whose weights are being compressed; it provides the reference outputs Y_ℓ used in layer-wise feature matching.

Calibration set

A small set of input sequences (typically WikiText-103 / C4, $B_{\text{tokens}} \sim 4096$) used to extract per-layer activations X_ℓ for the Hessian computation.

Block-sparsity

A sparsity pattern in which zero weights are organized into contiguous blocks of size $\geq B_{\text{block}}$, rather than scattered (unstructured) or in 2:4 semi-structured patterns.

2:4 **sparsity**

NVIDIA-Ampere-native semi-structured sparsity in which exactly 2 non-zero entries appear per contiguous group of 4. Invisible to the C SIMD engine.

Probe matrix $A \in \mathbb{R}^{K \times N}$

A small matrix used to predict, for each input x, which blocks should be active via $p(x) = \text{sign}(Ax) \in \{-1, +1\}^K$. Enables in-place zero-skip execution in the C engine.

Skippable fraction

The fraction of memory traffic that the C engine can avoid loading due to the sparsity pattern. Equals ≈ 0 for unstructured and 2:4, equals $\approx s$ for contiguous block sparsity.

B Proof Sketches for Main Theorems

B.1 Proof of Theorem 3.3 (ADMM Convergence)

The proof follows the standard KL-based framework for non-convex ADMM [6, 7, 8]. *Step 1 (sufficient descent).* Let $\mathcal{L}_\rho^{(t)} = \mathcal{L}_\rho(W'^{(t)}, Z_i^{(t)}, U_i^{(t)})$. The W'-update (6) minimizes a strongly convex quadratic (under Assumption 3.1), giving $\mathcal{L}_\rho^{(t+1)} - \mathcal{L}_\rho^{(t)} \leq -\frac{\mu+\rho_\Sigma}{2} \left\| W'^{(t+1)} - W'^{(t)} \right\|_F^2$. The Z_i-updates are projections onto non-expansive sets (under Assumption 3.2), giving $\mathcal{L}_\rho^{(t+1)} - \mathcal{L}_\rho^{(t)} \leq -\frac{\rho_i}{2} \left\| Z_i^{(t+1)} - Z_i^{(t)} \right\|_F^2$. *Step 2 (bounded dual residual).* The dual update (17) telescopes, and bounded iterates imply bounded duals. *Step 3 (KL inequality).* $\mathcal{L}_\rho$ is a semi-algebraic function (its components are polynomial and indicator-of-semi-algebraic-sets), hence satisfies the KL inequality with exponent $\theta \in [1/2, 1)$ [8]. By the standard KL-based argument, the iterates converge to a critical point $(W'^\star, Z_i^\star, U_i^\star)$ of $\mathcal{L}_\rho$, with rate $\left\| (W'^{(t)}, Z_i^{(t)}, U_i^{(t)}) - (W'^\star, Z_i^\star, U_i^\star) \right\| \leq \mathcal{O}(t^{-1/(2\theta-1)})$. For $\theta = 1/2$ (the generic case), this gives $\mathcal{O}(1/t)$. Integrating the descent inequality (**??**) over $t = 1, \ldots, T$ and using the convexity of $\mathcal{E}$ on the smooth part yields the ergodic rate (19).

B.2 Proof of Theorem 4.6 (Block-Sparsity Bound)

The proof has three parts. *(i) Coarsening penalty.* Block-sparsity selects $K_{\text{active}} = (1-s)K$ blocks; the dropped mass per block is the second-smallest block-magnitude. By Assumption 4.2, per-block energies are approximately uniform. The $\log B_{\text{block}}/B_{\text{block}}$ factor arises from the expected fraction of within-block mass lost when B_{block} entries are jointly thresholded vs. entry-wise; it is the standard coarsening penalty in combinatorial bandit literature. *(ii) Coupling*

penalty. The inter-block coupling η enters via the Schur-complement correction: the residual error in block k's solve propagates to neighbors through H_{kj}, inflating the error by $1+\eta/(1-\eta)$, which for $\eta = 0.2$ gives $1.25\times$ inflation. *(iii) Ternary quantization cost.* Entry-wise rounding (20) incurs ℓ_2 error of at most $S_k^2/4$ per entry; the off-diagonal Hessian deviation contributes the multiplicative factor. Summing yields (23).

B.3 Proof of Theorem 5.2 (Healed Drift Growth)

The proof tracks the drift through the three healing tiers. *(i) Tier 1.* Per-layer solves with corrected input $\tilde{X}_\ell$ reduce the per-layer residual to $\mathcal{O}(\sigma_{\text{probe}})$, where $\sigma_{\text{probe}} = \left\|\Delta_\ell X_\ell^{\text{donor}}\right\|_2 / \sqrt{B_{\text{block}}}$ is the per-block probe-alignment residual. The $\sqrt{L}$ scaling arises from the random-walk-like accumulation of these per-layer residuals, whose signs are approximately independent across layers due to the rotating eigenvector structure of J_ℓ. *(ii) Tier 2.* The 3-layer block solves at every $T_{\text{heal}} = 8$ layers provide a Schur-complement correction that reduces the drift accumulated over the previous 8 layers by a factor of $\sqrt{T_{\text{heal}}}$, converting linear growth $\mathcal{O}(L)$ into $\sqrt{L/T_{\text{heal}}} \cdot \sqrt{T_{\text{heal}}} = \sqrt{L}$ growth. *(iii) Tier 3.* The α healing neurons contribute the $\alpha L^2 \sigma_{\text{probe}}^2$ term: each healing neuron reduces residual by factor α but introduces σ_{probe}^2 second-order error. For $\alpha = 0.005$, $L = 80$, this term is bounded by 0.32. Summing the three contributions yields (29).

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